RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 2 (Lambda)
Topic(s): Complex Numbers and Sequences and Series (Solutions)

Source of Question: 9740/2012/I/Q3(modified)

1 A sequence $u_1, u_2, u_3, \ldots$ is given by

$$u_1 = 2 \text{ and } u_{n+1} = \frac{3u_n - 1}{6} \text{ for } n \geq 1.$$

(i) Find the exact values of u_2 and u_3, and the value of u_{10} to 5 decimal places. [3]

(ii) It is given that $u_n \to l$ as $n \to \infty$. Showing your working, find the exact value of l. [2]

(iii) Describe the behaviour of the sequence. [1]

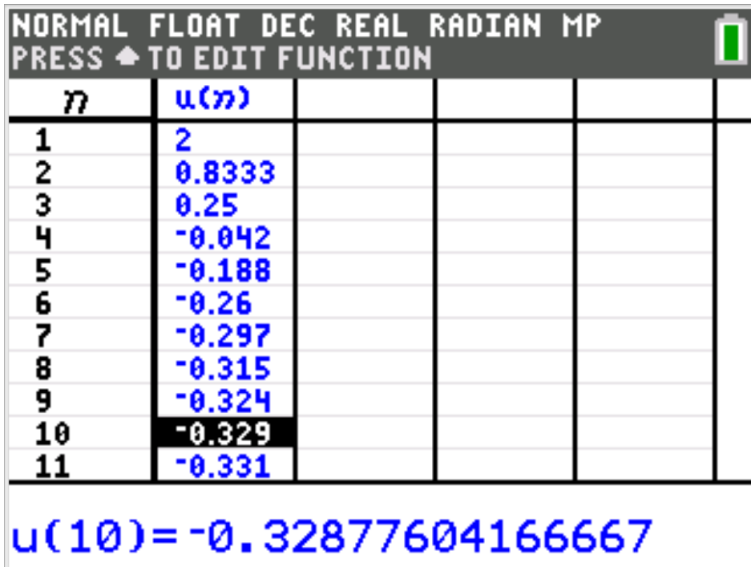
Solution:

(i)

$$u_2 = \frac{3(2)-1}{6} = \frac{5}{6}$$

$$u_3 = \frac{3\left(\frac{5}{6}\right)-1}{6} = \frac{1}{4}$$

From the GC, $u_{10} = -0.32878$ (5 dp)



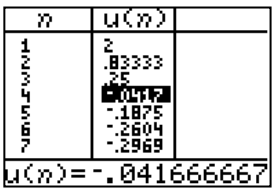
(ii)

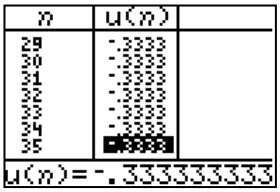
As $n \to \infty$, $u_n \to l$ and $u_{n+1} \to l$

We have $l = \frac{3(l)-1}{6} \Rightarrow 6l - 3l = -1 \Rightarrow l = -\frac{1}{3}$

(iii)

The sequence decreases and converges to $-\frac{1}{3}$.





Source of Question: TJC Prelim 9758/2020/02/Q1

2 A sequence of numbers $u_1, u_2, u_3, \ldots$ has a sum S_n where $S_n = \sum_{r=1}^{n} u_r$.

(i) It is known that the sum of the first three terms of the sequence is 17 and $\sum_{n=1}^{3} S_n = 30$.

Furthermore, the third term of the sequence is twice the first term of the sequence. Find the first three terms of the sequence. [3]

(ii) A constant k is subtracted from each term of the sequence so that the first three terms of the modified sequence are now three consecutive terms of a geometric progression with a common ratio greater than 1.
Find the value of k and the common ratio of the geometric progression. [3]

Solution:

(i)	$u_1 + u_2 + u_3 = 17$ --- (1) $S_1 + S_2 + S_3 = 30$ $u_1 + (u_1 + u_2) + (u_1 + u_2 + u_3) = 30$ $3u_1 + 2u_2 + u_3 = 30$ --- (2) $u_3 = 2u_1 \Rightarrow 2u_1 - u_3 = 0$ --- (3) From GC, first 3 terms of sequence are $u_1 = 4$, $u_2 = 5$ and $u_3 = 8$.
(ii)	After subtracting k, the three terms are consecutive terms of a GP and hence $\frac{8-k}{5-k} = \frac{5-k}{4-k}$ $\Rightarrow 32 - 12k + k^2 = 25 - 10k + k^2$ $\Rightarrow k = \frac{7}{2}$ $\Rightarrow r = \frac{8-\frac{7}{2}}{5-\frac{7}{2}} = 3$

Source of Question: CJC Prelim 9758/2022/II/Q2 (modified)

3 A sequence $a_1, a_2, a_3, \ldots$ is such that $a_n = 2a_{n-1} - 3n + K$, where K is a constant, and $n \geq 2$.

(i) Given that $a_1 = 2$ and $a_2 = 4$, find K, a_3 and a_{18}. [2]

It is known that the nth term of this sequence is given by $a_n = p(2^n) + qn + r$, where p, q and r are constants.

(ii) Find p, q and r. [3]

(iii) Find $\sum_{n=1}^{N} a_n$ in terms of N. [2]

Solution:

(i)

Given that $a_1 = 2$ and $a_2 = 4$,

$4 = 2(2) - 3(2) + K \quad \Rightarrow \quad K = 6$

Using GC, $a_3 = 5$

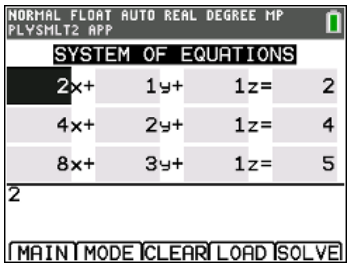
$a_{18} = -131018$

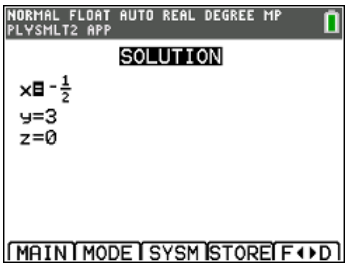
(ii)

Since $a_n = p(2^n) + qn + r$,

$$\left.\begin{aligned} a_1 = 2 \ &\Rightarrow \ p(2^1) + q(1) + r = 2 \\ a_2 = 4 \ &\Rightarrow \ p(2^2) + q(2) + r = 4 \\ a_3 = 5 \ &\Rightarrow \ p(2^3) + q(3) + r = 5 \end{aligned}\right\}$$

Solving this system of linear equations :





$\therefore \ p = -\frac{1}{2}$, $q = 3$, and $r = 0$.

$\therefore \ a_n = -\frac{1}{2}(2^n) + 3n$

(iii)

$$a_n = -\frac{1}{2}(2^n) + 3n$$

$$\sum_{n=1}^{N} a_n = \sum_{n=1}^{N} -\frac{1}{2}(2^n) + 3n$$

$$= -\frac{1}{2}\sum_{n=1}^{N} 2^n + 3\sum_{n=1}^{N} n$$

$$= -\frac{1}{2}\left(2^1 + 2^2 + 2^3 + \ldots + 2^N\right) + 3\left(1 + 2 + 3 + \ldots + N\right)$$

$$= -\frac{1}{2}\left(\frac{2^1(2^N - 1)}{2 - 1}\right) + 3\left(\frac{N}{2}(1 + N)\right)$$

$$= (1 - 2^N) + \frac{3}{2}N(1 + N)$$

Source of Question: CJC Prelim 9740/2015/I/Q5(modified)

4 A sequence $u_1, u_2, u_3, ...$ is defined by the following recurrence relation,

$$u_1 = 1, \quad u_{n+1} = \frac{1}{2}u_n + 1 \text{ for all } n \in \mathbb{Z}^+.$$

Another sequence $v_1, v_2, v_3, ...$ is defined by $v_n = u_n - 2$ for all $n \in \mathbb{Z}^+$.

(i) Find a recurrence relation of the form $v_{n+1} = \mathrm{f}(v_n)$. Hence show that the sequence $v_1, v_2, v_3, ...$ is a geometric progression with common ratio $\frac{1}{2}$. [3]

(ii) Find the limit of the sequence $v_1, v_2, v_3, ...$ and that of the sequence $u_1, u_2, u_3, ...$ when $n \to \infty$. [3]

(ii) Find the sum to infinity of the series $v_1 + v_2 + v_3 ....$ [1]

(iv) Find the sum of the first n terms of the sequence $u_1, u_2, u_3, ...$ Determine whether the series $u_1 + u_2 + u_3 + ...$ converges. [3]

Solution:

(i)

$$v_{n+1} + 2 = \frac{1}{2}(v_n + 2) + 1$$

$$v_{n+1} + 2 = \frac{1}{2}v_n + 1 + 1$$

$$v_{n+1} = \frac{1}{2}v_n$$

Since $\frac{v_{n+1}}{v_n} = \frac{1}{2} = \text{constant}$, $v_1, v_2, v_3, ...$ is a geometric progression with common ratio $\frac{1}{2}$.

(ii)

$$v_1 = u_1 - 2 = -1$$

$$v_n = (-1)\left(\frac{1}{2}\right)^{n-1}$$

When $n \to \infty$, $v_n = (-1)\left(\frac{1}{2}\right)^{n-1} \to 0$ and thus the limit of sequence $v_1, v_2, v_3, ...$ is 0.

Since $u_n = v_n + 2$, when $n \to \infty, u_n \to 2$ and the limit of sequence $u_1, u_2, u_3, ...$ is 2.

(iii)

$$\text{Sum to infinity} = \frac{-1}{1 - \frac{1}{2}} = -2$$

(iv)

$$\sum_{r=1}^{n} v_r = \frac{(-1)\left(1 - \left(\frac{1}{2}\right)^n\right)}{1 - \frac{1}{2}} = -2\left(1 - \left(\frac{1}{2}\right)^n\right) = 2\left(\frac{1}{2}\right)^n - 2$$

$$\sum_{r=1}^{n} u_r = \sum_{r=1}^{n}(v_r + 2) = \sum_{r=1}^{n} v_r + 2n = 2\left(\frac{1}{2}\right)^n - 2 + 2n$$

	When $n \to \infty$, $2\left(\frac{1}{2}\right)^n \to 0$ and $2n \to \infty$. Thus $\sum_{r=1}^{n} u_r \to \infty$, i.e the series $u_1 + u_2 + u_3 + ...$ does not converge.

Source of Question: MI Prelim 9758/2021/I/Q11

5 Bank *A* offers a study loan to students enrolled in an undergraduate course of study. The key features of the loan are:

- interest-free during the course of study,
- fixed monthly interest of 0.3% upon graduation,
- minimum monthly repayment of \$100.

Ali decides to take a study loan of \$50000 from Bank *A* on 1 January 2021 at the start of his 3-year undergraduate course.

(a) During his course of study, Ali pays \$200 at the end of January 2021 and on the last day of each subsequent month, he pays \$10 more than in the previous month. Thus on 28 February 2021, he pays \$210 and on 31 March 2021, he pays \$220, and so on. How much does Ali owe the bank at the end of his course of study? [2]

(b) Bank *A* charges interest on any outstanding amount of the loan on the first day of each month, starting on the month right after a student graduates.

(i) Upon graduation, Ali immediately found a job and decides to pay \$900 to bank *A* on the last day of each month, starting on the month right after he graduated. Show that Ali owes the bank \$34121 (to the nearest dollar) on the last day of the 3rd month after he graduated. [2]

(ii) Use the formula for the sum of a geometric progression to find an expression for the amount owed by Ali on the last day of the *n*th month after he graduated. Hence find in which month Ali pays off his study loan. [5]

(iii) Find the total amount of interest that Ali paid. [2]

(iv) If Ali decides to pay off his study loan within 3 years upon graduation, i.e. at the end of December 2026, what is the minimum amount, to the nearest dollar, that he needs to pay per month after graduation? [3]

Solution:

(a)	Amount of money Ali paid at the end of 3 years $= \frac{36}{2}\left[2(200) + (36-1)(10)\right]$ $= 13500$

Amount Ali owes the bank at the end of 3 years
$= 50000 - 13500$
$= 36500$

(b)(i)

At the end of 1 month, amount owed
$= 1.003(36500) - 900$
$= 35709.50$

At the end of 2 months, amount owed
$= 1.003(35709.50) - 900$
$= 34916.6285$

At the end of 3 months, amount owed
$= 1.003(34916.6285) - 900$
$= 34121.38$
$= 34121$ (nearest dollar)

(ii)

Month	Amount owed at the start of month	Amount owed at the end of month
1	$1.003(36500)$	$1.003(36500) - 900$
2	$1.003\left[1.003(36500) - 900\right]$ $= 1.003^2(36500) - 1.003(900)$	$1.003^2(36500) - 1.003(900)$ -900
3	$1.003\left[1.003^2(36500) - 1.003(900) - 900\right]$ $= 1.003^3(36500) - 1.003^2(900) - 1.003(900)$	$1.003^3(36500) - 1.003^2(900) - 1.003(900)$ -900
…	…	…
n		$1.003^n(36500) - 1.003^{n-1}(900)$ $-1.003^{n-2}(900) - \ldots - 900$

On the last day of the nth month, Ali owed

$$1.003^n(36500) - 1.003^{n-1}(900) - 1.003^{n-2}(900) - \ldots - 900$$
$$= 1.003^n(36500) - \left[900 + 1.003(900) \ldots + 1.003^{n-2}(900) + 1.003^{n-1}(900)\right]$$
$$= 1.003^n(36500) - 900\left(1 + 1.003 + \ldots + 1.003^{n-2} + 1.003^{n-1}\right)$$
$$= 1.003^n(36500) - 900\left[\frac{1.003^n - 1}{1.003 - 1}\right]$$
$$= 1.003^n(36500) - 300000\left(1.003^n - 1\right)$$

When Ali pays off his study loan,

$$1.003^n(36500) - 300000\left(1.003^n - 1\right) \leq 0$$

	Using GC, n | $1.003^n(36500)-300000(1.003^n-1)$ 43 | 276.54 44 | −622.6 45 | −1524 Ali takes 44 months to pay off his study loan, i.e. he pays off his study loan in August 2027.
(iii)	**Method 1:** Since Ali pays $\$(1.003\times 276.54)$ in the last month, total interest Ali paid $=43\times 900+(1.003\times 276.54)-36500$ $=2477.37$ (nearest cent) **Method 2:** Since Ali pays \$276.54 in the last month, total amount he paid $=13500+43\times 900+(1.003\times 276.54)$ $=52477.37$ Total amount of interest Ali paid $=52477.37-50000$ $=2477.37$ (nearest cent)
(iv)	Let the amount that Ali pays per month upon graduation be \$$x$. $1.003^{36}(36500)-x\left[\dfrac{1.003^{36}-1}{1.003-1}\right]\le 0$ $40656.16902-37.956x\le 0$ $x\ge 1071.14$ Ali needs to pay \$1072 per month.

Source of Question: EJC Prelim 9758/2019/I/Q3

6 A geometric progression has first term a and common ratio r, and an arithmetic progression has first term a and common difference d, where a and d are non-zero. The sums of the first 2 and 4 terms of the arithmetic progression are equal to the respective sums of the first 2 and 4 terms of the geometric progression.

(i) By showing $r^3+r^2-5r+3=0$, or otherwise, find the value of the common ratio. [5]

(ii) Given that $a<0$ and the nth term of the geometric progression is positive, find the smallest possible value of n such that the nth term of the geometric progression is more than 1000 times the nth term of the arithmetic progression. [3]

Solution:

(i)

GP: a, ar, ar^2, ar^3

AP: $a, a+d, a+2d, a+3d$

Since the sum of the first 2 terms of the AP is equal to the sum of the first 2 terms of the GP, and they both have a as first term, then their 2nd terms have the same value:

$ar = a + d \quad \cdots(1)$

Since the sums of the first 2 and 4 terms of the AP are equal to the respective sums of the first 2 and 4 terms of the GP, then the sum of the of the 3rd and 4th terms of the AP is the same as that of the GP: $ar^2 + ar^3 = a + 2d + a + 3d \quad \cdots(2)$

Put (1) in (2): $ar^2 + ar^3 = 2a + 5(ar - a)$

Since $a \neq 0$, $r^2 + r^3 = 2 + 5r - 5$

$r^2 + r^3 - 5r + 3 = 0$

Using GC, $r = -3$. $r \neq 1$, otherwise $d = 0$ from equation (1)

Alternative solution

$$2a + d = \frac{a(r^2 - 1)}{r - 1} \text{ -- (1)}$$

$$4a + 6d = \frac{a(r^4 - 1)}{r - 1} \text{ -- (2)}$$

6 x (1)-2:

$$8a = \frac{6a(r^2 - 1)}{r - 1} - \frac{a(r^4 - 1)}{r - 1}$$

$$8a(r - 1) = a(6r^2 - 6 - r^4 + 1)$$

$$8r - 8 = 6r^2 - r^4 - 5$$

$$r^4 - 6r^2 + 8r - 3 = 0$$

Solving:

$r = -3$ or 1 (rej)

(ii)

$a(-3)^{n-1} > 1000[a + (n-1)d]$

Note: $d = a(-3+1) - 2a = -4a$

$a(-3)^{n-1} > 1000[a + (n-1)(-4a)]$

$a(-3)^{n-1} > 1000a(5 - 4n)$

Since $a < 0$,

$(-3)^{n-1} < 1000(5 - 4n)$

Since n^{th} term of GP is positive, i.e. $a(-3)^{n-1} > 0$

$(-3)^{n-1}$ is negative $\Rightarrow n$ is even

	From GC (table) n | $(-3)^{n-1}$ | $1000(5-4n)$ 10 | $-19683 >$ | -35000 12 | $-177147 <$ | -43000 Smallest $n=12$

Source of Question: NYJC Prelim 9758/2019/II/Q3

7 When a ball is dropped from a height of H m above the ground, it will rebound to a height of eH m where $0<e<1$. The height of each successive bounce will be e times of that of its previous height. It is also known that the time taken between successive bounce is given by $t=0.90305\sqrt{h}$ where h is the maximum height of the ball from the ground between these bounces. We can assume that there is negligible air resistance.

A ball is now dropped from a height of 10 m from the ground. Let t_n be the time between the nth and $(n+1)^{\text{th}}$ bounce.

(i) Show that the total distance travelled by the ball just before the nth bounce is $\dfrac{10(1+e-2e^n)}{1-e}$. [3]

(ii) Show that t_n is a geometric sequence. State the common ratio for this sequence. [3]

(iii) Find in terms of e the total distance the ball will travel and the time taken when it comes to rest. You may assume that between any two bounces, the time taken for the ball to reach its maximum height is the same as the time it takes to return to the ground. [3]

Solution:

(i)	Let S_n be the total distance travelled by the ball just before the n-th bounce. Thus $S_n = 10+2(10e)+2(10e^2)+\cdots+2(10e^{n-1})$ $= 20+20e+20e^2+\cdots+20e^{n-1}-10$ $= \dfrac{20(1-e^n)}{1-e}-10$ $= \dfrac{10(1+e-2e^n)}{1-e}$
(ii)	Let d_k be the maximum height of the ball after the k-th bounce. Thus $d_k=10e^k$. Hence $t_k=0.90305\sqrt{d_k}$. Thus for $k\in\mathbb{Z}^+$, $\dfrac{t_{k+1}}{t_k}=\dfrac{0.90305\sqrt{d_{k+1}}}{0.90305\sqrt{d_k}}$ $=\dfrac{\sqrt{10e^{k+1}}}{\sqrt{10e^k}}=\sqrt{e}$ Hence t_n is a geometric sequence with common ratio $\sqrt{e}$.

(iii)	As $n \to \infty$, the ball will come to rest. Thus total distance travelled is $S = \lim_{n\to\infty}\left(\frac{10(1+e-2e^n)}{1-e}\right)$ $= \frac{10(1+e)}{1-e}$ Total time taken $= 0.5(0.90305)\sqrt{10} + \sum_{n=1}^{\infty} t_n$ $= 1.4278 + \frac{0.90305\sqrt{10e}}{1-\sqrt{e}}$ $= 1.43 + \frac{2.86\sqrt{e}}{1-\sqrt{e}}$

Source of Question: NJC JC2 CT2 9758/2017/P1/Q6

8 Do not use a calculator in answering this question.

(i) Solve the equation $z^2 + 1 + \left(2\sqrt{2}\right)\mathrm{i} = 0$, leaving the roots in the form $x + \mathrm{i}y$, where x and y are real. [4]

(ii) One root of the equation $3s^3 - 8s^2 + 31s - 18 = 0$ is $1 + \left(2\sqrt{2}\right)\mathrm{i}$. Find the other roots. [3]

Deduce the exact roots of the equation $3p^6 + 8p^4 + 31p^2 + 18 = 0$. [4]

Solution:

(i)	Let $z = x + \mathrm{i}y$. $(x+\mathrm{i}y)^2 + 1 + \left(2\sqrt{2}\right)\mathrm{i} = 0$ $\Rightarrow x^2 + \mathrm{i}2xy - y^2 + 1 + \left(2\sqrt{2}\right)\mathrm{i} = 0$ Comparing real and imaginary parts, $x^2 - y^2 + 1 = 0$ -----(1) $2xy + \left(2\sqrt{2}\right) = 0$ -----(2) From (2): $y = -\frac{\sqrt{2}}{x}$ Substituting into (1): $x^2 - \left(-\frac{\sqrt{2}}{x}\right)^2 + 1 = 0$

	$\Rightarrow x^4 + x^2 - 2 = 0$ $\Rightarrow x^2 = \dfrac{-1 \pm \sqrt{1+8}}{2}$ $\Rightarrow x^2 = \dfrac{-1 \pm 3}{2}$ $\Rightarrow x^2 = 1$ or $x^2 = -2$ $\left(\text{N.A. } \because x^2 \geq 0\right)$ $\Rightarrow x = 1$ or -1 If $x = 1,$ then $y = -\sqrt{2}.$ If $x = -1,$ then $y = \sqrt{2}.$ $\therefore$ the roots are $z = 1 - \sqrt{2}\text{i}$ and $z = -1 + \sqrt{2}\text{i}.$
(ii)	Since the equation has real coefficients and constant, by the conjugate root theorem, $s = 1 - \left(2\sqrt{2}\right)\text{i}$ is also a root. $3s^3 - 8s^2 + 31s - 18 = \left(s - 1 - \left(2\sqrt{2}\right)\text{i}\right)\left(s - 1 + \left(2\sqrt{2}\right)\text{i}\right)(3s - A)$ $= \left((s-1)^2 - \left(2\sqrt{2}\text{i}\right)^2\right)(3s - A)$ $= \left(s^2 - 2s + 1 + 8\right)(3s - A)$ $= \left(s^2 - 2s + 9\right)(3s - A)$ Compare the constant term: $-18 = -9A \Rightarrow A = 2$ Thus, the other roots are $s = 1 - \left(2\sqrt{2}\right)\text{i}$ and $s = \dfrac{2}{3}.$
(iii)	Replace s by $-p^2$ in the equation in (ii). $3\left(-p^2\right)^3 - 8\left(-p^2\right)^2 + 31\left(-p^2\right) - 18 = 0$ $\Rightarrow -3p^6 - 8p^4 - 31p^2 - 18 = 0$ $\Rightarrow 3p^6 + 8p^4 + 31p^2 + 18 = 0$ From (ii), $-p^2 = 1 + \left(2\sqrt{2}\right)\text{i} \Rightarrow p^2 + 1 + \left(2\sqrt{2}\right)\text{i} = 0.$ From (i), the roots are $p = 1 - \sqrt{2}\text{i}$ and $p = -1 + \sqrt{2}\text{i}.$ By the conjugate root theorem, the other 2 roots are $p = 1 + \sqrt{2}\text{i}$ and $p = -1 - \sqrt{2}\text{i}.$ Also, $-p^2 = \dfrac{2}{3} \Rightarrow p = \pm\sqrt{-\dfrac{2}{3}}$ $\Rightarrow p = \left(\sqrt{\dfrac{2}{3}}\right)\text{i}$ or $\left(-\sqrt{\dfrac{2}{3}}\right)\text{i}$

Source of Question: NJC Prelim 9758/2022/01/Q4(i)

9 Two complex numbers w and u satisfy the equations

$$w^* - 2\mathrm{i}u = 8 \quad \text{and} \quad (2\mathrm{i}-1)w + 2u^* = 4.$$

Find w and u, giving your answers in the form $x + \mathrm{i}y$, where x and y are real. [4]

Hence find $|u^* - w|$ and $\arg(u^* - w)$. [2]

Solution:

Method 1

$$w^* - 2\mathrm{i}u = 8 \quad \Rightarrow \quad w^* = 8 + 2\mathrm{i}u \quad \Rightarrow \quad w = 8 - 2\mathrm{i}u^* \quad \cdots(1)$$

$$(2\mathrm{i}-1)w + 2u^* = 4 \quad \cdots(2)$$

Put (1) in (2): $(2\mathrm{i}-1)(8 - 2\mathrm{i}u^*) + 2u^* = 4$

$$16\mathrm{i} - 8 + 4u^* + 2\mathrm{i}u^* + 2u^* = 4$$

$$(6 + 2\mathrm{i})u^* = 12 - 16\mathrm{i}$$

$$u^* = \frac{12 - 16\mathrm{i}}{6 + 2\mathrm{i}} = 1 - 3\mathrm{i} \quad \text{from GC}$$

$$u = 1 + 3\mathrm{i}$$

From (1), $w = 8 - 2\mathrm{i}u^* = 8 - 2\mathrm{i}(1 - 3\mathrm{i}) = 2 - 2\mathrm{i}$

Method 2

$$w^* - 2\mathrm{i}u = 8 \quad \Rightarrow \quad \mathrm{i}w^* + 2u = 8\mathrm{i} \quad \Rightarrow \quad -\mathrm{i}w + 2u^* = -8\mathrm{i} \quad \cdots(1)$$

$$(2\mathrm{i}-1)w + 2u^* = 4 \quad \cdots(2)$$

(2) – (1): $(3\mathrm{i}-1)w = 4 + 8\mathrm{i}$

$$w = \frac{4 + 8\mathrm{i}}{3\mathrm{i} - 1} = 2 - 2\mathrm{i} \quad \text{from GC}$$

From $\mathrm{i}w^* + 2u = 8\mathrm{i}$, $u = \dfrac{8\mathrm{i} - \mathrm{i}w^*}{2}$

$$= \frac{8\mathrm{i} - \mathrm{i}(2 + 2\mathrm{i})}{2} = 1 + 3\mathrm{i}$$

$|u^* - w| = |1 - 3\mathrm{i} - 2 + 2\mathrm{i}| = |-1 - \mathrm{i}| = \sqrt{2}$

$\arg(u^* - w) = \arg(-1 - \mathrm{i}) = -\frac{3\pi}{4}$

Source of Question: JJC JC2 Mid-Year CT 9758/2018/01/Q4(iii)(iv)

10 Given $u = 1+\sqrt{3}\mathrm{i}$ and $v = 1-\mathrm{i}$ are roots of the equation $z^4 + az^3 + bz^2 + cz + d = 0$, where a, b, c and $d \in \mathbb{R}$,

(a) find the values of a, b, c and d. [4]

The complex numbers $\mathrm{i}u$ and v^* are represented by the points P and Q respectively on an Argand diagram.

(b) Find the area of triangle OPQ. [3]

Solution:

(a)

Since $a, b, c, d \in \mathbb{R}$, by Complex Conjugate Root Theorem, u^* and v^* are also roots of the equation $z^4 + az^3 + bz^2 + cz + d = 0$.

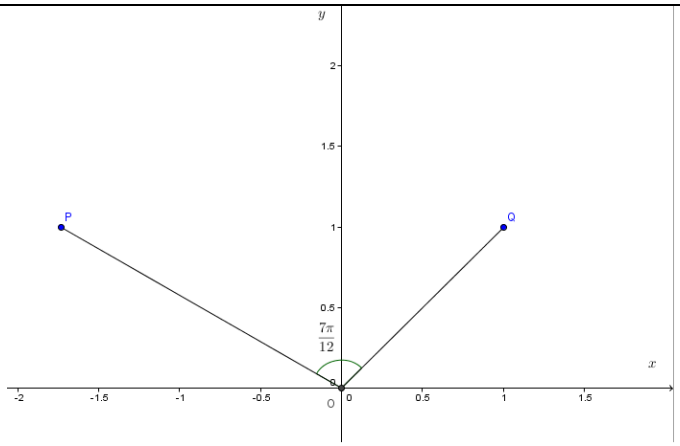
$$\Rightarrow \left(z-\left(1+\sqrt{3}\mathrm{i}\right)\right)\left(z-\left(1-\sqrt{3}\mathrm{i}\right)\right)\left(z-\left(1-\mathrm{i}\right)\right)\left(z-\left(1+\mathrm{i}\right)\right)=0$$

$$\Rightarrow \left(z^2-2z+4\right)\left(z^2-2z+2\right)=0$$

$$\Rightarrow z^4-4z^3+10z^2-12z+8=0$$

$$\therefore a=-4,\ b=10,\ c=-12,\ d=8$$

(b)



Argument of $\mathrm{i}u = \dfrac{\pi}{3}+\dfrac{\pi}{2}=\dfrac{5\pi}{6}$

Argument of $v^* = \dfrac{\pi}{4}$

$$\Rightarrow \angle POQ = \frac{5\pi}{6}-\frac{\pi}{4}=\frac{7\pi}{12}$$

Area of traingle $POQ = \dfrac{1}{2}\times 2\times\sqrt{2}\times\sin\left(\dfrac{7\pi}{12}\right)$

$= 1.37$ units^2

Source of Question: NJC Prelim 9758/2025/P2/Q1

11 Do not use a calculator in answering this question.

The complex number w is such that $w = \mathrm{i} - \sqrt{3}$.

(i) Find $|w|$ and $\arg(w)$ in exact form. [2]

(ii) Represent w, $-w$ and $2-w$ in the same Argand diagram, illustrating clearly the geometrical relationship between them. [3]

(iii) Hence, find $\arg(2-w)$ exactly. [2]

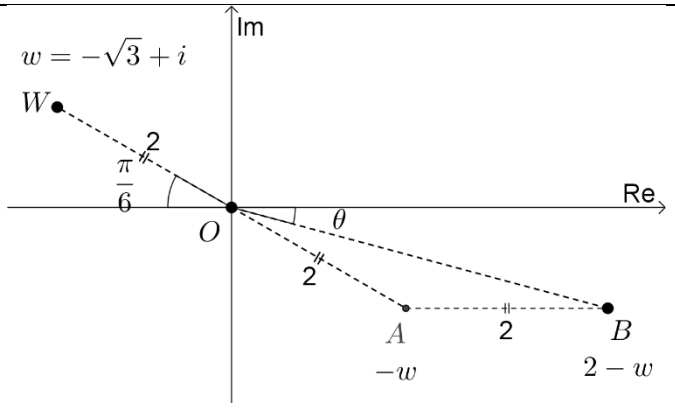
Solution:

(i)

$$|w| = \left|\mathrm{i} - \sqrt{3}\right| = \sqrt{\left(-\sqrt{3}\right)^2 + 1^2} = 2$$

$$\arg(w) = \pi - \tan^{-1}\frac{1}{\sqrt{3}} = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

(ii)



The point W represents w and the point B represents $2-w$.

(iii)

Since $|w| = 2$, the triangle OAB is an isosceles triangle.

AB is parallel to the real-axis, so $\angle AOB = \angle ABO = \theta$.

$$2\theta = \frac{\pi}{6}$$

$$\theta = \frac{\pi}{12}$$

$$\arg(2-w) = -\frac{\pi}{12}.$$

Source of Question: YIJC Prelim 9758/2025/P2/Q2 (modified)

12 Do not use a calculator in answering this question.

It is given that $z = \sqrt{3} - \mathrm{i}$ and $w = 2\mathrm{i}z$.

(a) Find the modulus and argument of z. [2]

O, A, B and C represent the complex numbers 0, z, $z+w$ and w respectively.

(b) Illustrate these four points on a single diagram. [2]

(c) Deduce the modulus and argument of w. [2]

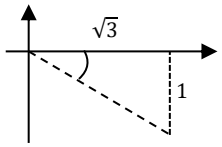
(d) Identify the shape of the quadrilateral $OABC$ and state its area. [2]

Solution:

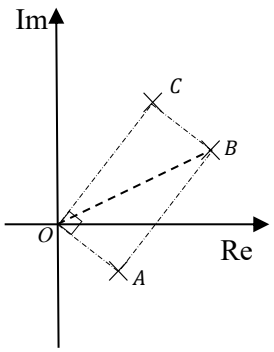
(a)

$$|z| = \sqrt{\left(\sqrt{3}\right)^2 + 1^2} = 2$$

$$\arg(z) = -\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$$



(b)



(c) Using the geometrical relationship between z & iz, and referring to the diagram,

$$|w| = 2|\mathrm{i}z| = 2|z| = 2 \times 2 = 4$$

$$\arg(w) = \arg(z) + \frac{\pi}{2} = -\frac{\pi}{6} + \frac{\pi}{2} = \frac{\pi}{3}$$

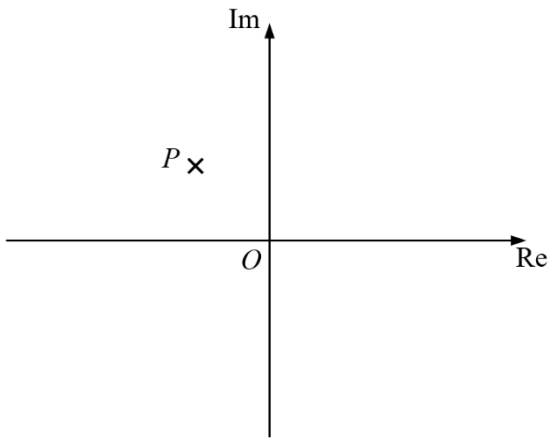
(d) Since O, A, B, C forms a parallelogram and $\angle AOC = \frac{\pi}{2}$, thus $OABC$ is a **rectangle**.

Area of rectangle = length $\times$ breadth $= 2 \times 4 = 8$.

Source of Question: VJC Prelim 9758/2025/P2/Q4

13 Do not use a calculator in answering this question.

The complex number z has modulus 1 and argument θ, where $\frac{\pi}{2} < \theta < \pi$, and the complex number w is given by $w = \mathrm{i}\sqrt{3}z$. The point P on the Argand diagram represents z.



(a) On the copy of the Argand diagram with origin O in the Printed Answer Booklet, plot the points Q and R to represent w and $z - w$ respectively. Show clearly the geometrical relationship between the points P, Q and R. [3]

(b) Find the area of the quadrilateral $ORPQ$. [1]

It is given that $\theta = \frac{3\pi}{4}$.

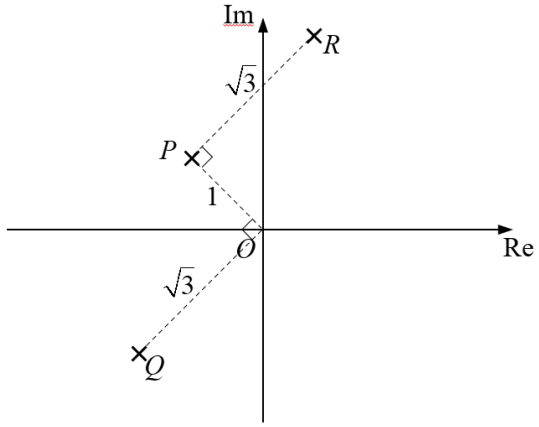
(c) Find z in the form $x + y\mathrm{i}$, where x and y are real numbers. [2]

(d) Show that $z - w = k\left[\left(\sqrt{3}-1\right)+\left(\sqrt{3}+1\right)\mathrm{i}\right]$, where k is a constant to be determined. [2]

(e) Hence show that $\tan\frac{5\pi}{12} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$. [1]

Solution:

(a)



(b) Area $ORPQ = \frac{1}{2}(1)\left(\sqrt{3}+\sqrt{3}\right) = \sqrt{3}$

(c)

$$\text{Length } x = 1\cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\text{Length } y = 1\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

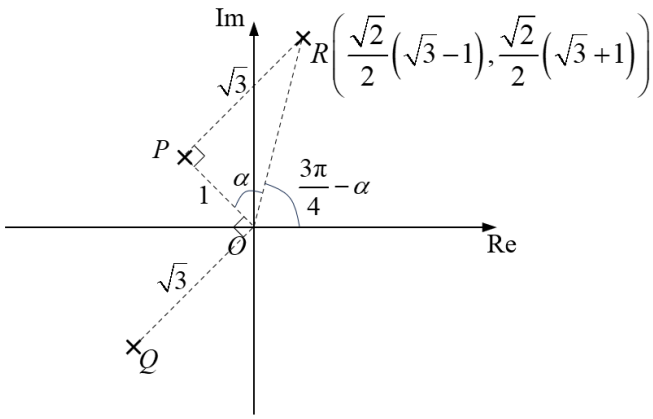
$$\therefore z = -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\text{i}$$

(d)

$$\begin{aligned} z - w &= \left(z - \sqrt{3}\text{i}z\right) \\ &= \left(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\text{i}\right)\left(1 - \sqrt{3}\text{i}\right) \\ &= \frac{1}{\sqrt{2}}\left(-1 + \text{i}\right)\left(1 - \sqrt{3}\text{i}\right) \\ &= \frac{1}{\sqrt{2}}\left(-1 + \sqrt{3} + \text{i} + \sqrt{3}\text{i}\right) \\ &= \frac{1}{\sqrt{2}}\left(\left(\sqrt{3} - 1\right) + \left(\sqrt{3} + 1\right)\text{i}\right) \end{aligned}$$

$$k = \frac{1}{\sqrt{2}}$$

(e)



$$\alpha = \tan^{-1}\frac{\sqrt{3}}{1} = \frac{\pi}{3}$$

$$\tan\left(\frac{3\pi}{4} - \frac{\pi}{3}\right) = \frac{\frac{\sqrt{2}}{2}\left(\sqrt{3}+1\right)}{\frac{\sqrt{2}}{2}\left(\sqrt{3}-1\right)}$$

$$\tan\frac{5\pi}{12} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$$